2 Million US Wildfires:
Hotspot Identification and Fire-spread Warning Model

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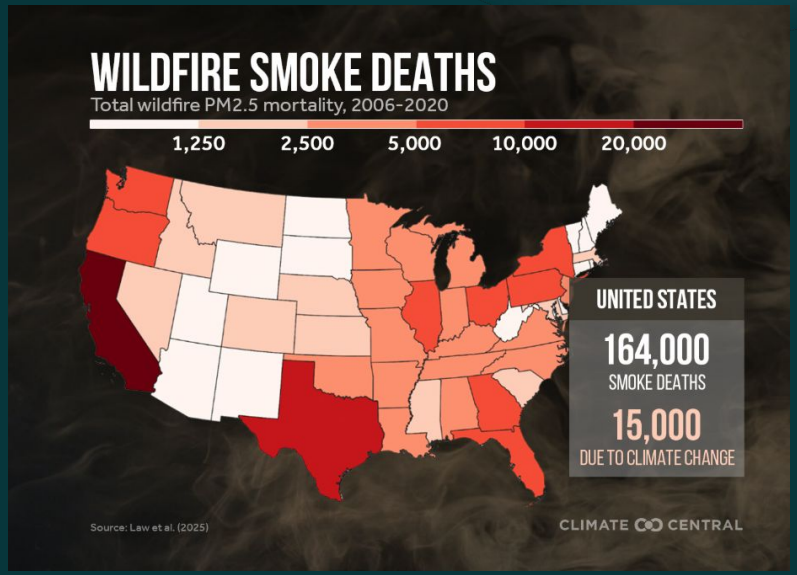
Introduction & Purpose

- Wildfire effects
  - Loss of life, economic damage, impact on health
- Identify high-risk regions
  - K-means
- Predict which fires are likely to spread
  - Decision Tree Classification



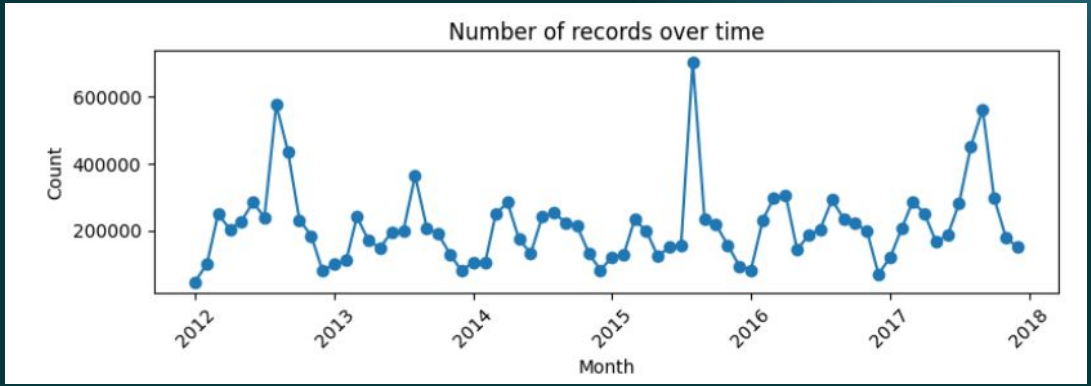
WildfireDB

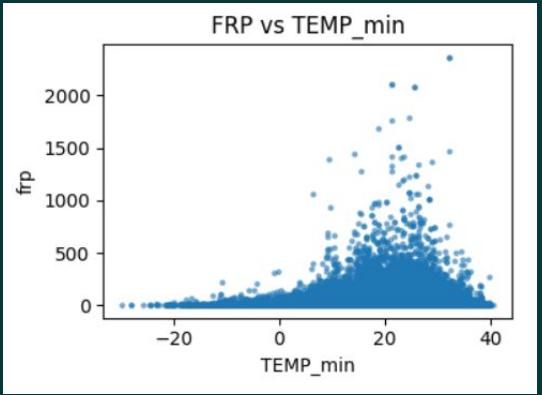
- Open-source temporal dataset for wildfire research
- Combined various data sources on:
  - Fire detection
  - Vegetation
  - Topography
  - Weather
- 17+ million entries, 140+ features
- From 2012-2018
- Uses NASA’s VIIRS fire detection product

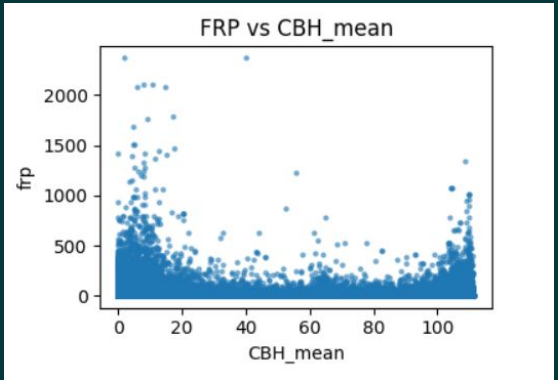
Data Exploration

- Areas affected by Wildfire can vary by season & temperature

- Vegetation features contribute to fire intensity







Note: EDA conducted on sampled data

Preprocessing

- Remove entries with commas (columns OR rows)
- Remove descriptive columns (“Shape”, ”Neighbour_Shape”, ”Polygon_ID”, “Acquisition_time”)
- Missing value handling: mean imputation

```
from pyspark.ml.feature import Imputer

input_cols = [c for c in df.columns if c != "Neighbour_frp"]

imputer = Imputer(
    inputCols=input_cols,
    outputCols=input_cols
).setStrategy("mean")

model = imputer.fit(df)
df = model.transform(df)
```

Methodology - Algorithms

K-means clustering:

- This algorithm divides the data into a given number of clusters by iteratively computing each cluster's mean and re-grouping the data.
- We selected this algorithm due to its efficiency and its applicability to the problem of hotspot identification.

Decision Trees:

- This algorithm exploits highly predictive columns and splits in the data in order to produce an accurate series of choices that produce a binary decision.
- We selected decision trees since our second problem involved binary classification and we had a large number of possibilities for predictive columns.

Methodology – K-means Pipeline

- Type casting, removing interval-like entries
- Select only FRP and Location features
- Scaling FRP only
- Final DF for k-means includes
  - FRP
  - Latitude
  - Longitude
  - FRP Vector
  - Scaled FRP
  - Features Vector (K-means input)

```
filtered_df = df.filter(~col("frp").contains(","))
filtered_df = filtered_df.withColumn("frp", df["frp"].cast(FloatType()))
features_df = filtered_df.select("frp", "c_latitude","c_longitude")
```

```
frp_assembler = VectorAssembler(
    inputCols=["frp"],
    outputCol="frp_vec"
)
frp_vector = frp_assembler.transform(features_df)
scaler = StandardScaler(
    inputCol="frp_vec",
    outputCol="scaled_frp",
    withMean=True,
    withStd=True
)
scaled_model = scaler.fit(frp_vector)
scaled_data = scaled_model.transform(frp_vector)
```

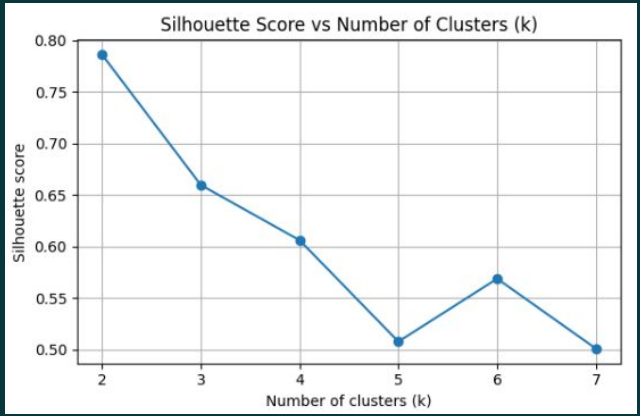
Methodology - Decision Tree Pipeline

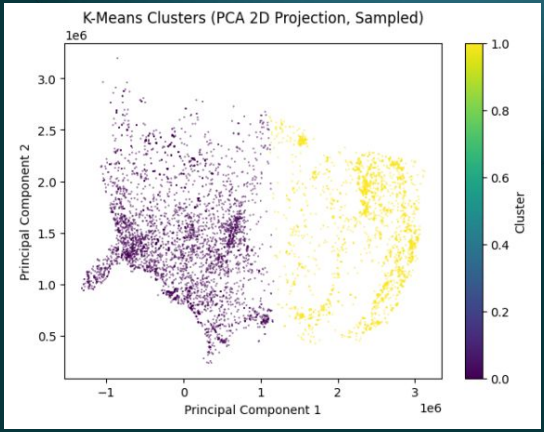
- Turn NULLs in “Neighbour_frp” column to 0.0 and anything else to 1.0
- Add weight column (inverse of representation percentage) to combat class imbalance
- Generate feature vector
- Determine ideal “max_depth” using a subset of the data
- Run decision tree algorithm on full dataset
- Evaluate using accuracy and F1 score

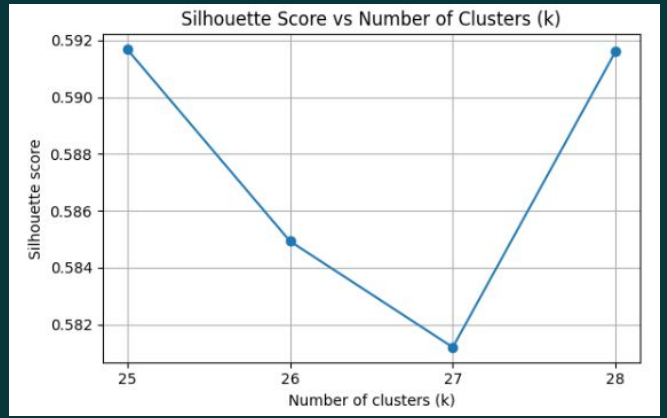
```
from pyspark.ml.classification import DecisionTreeClassifier
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
```

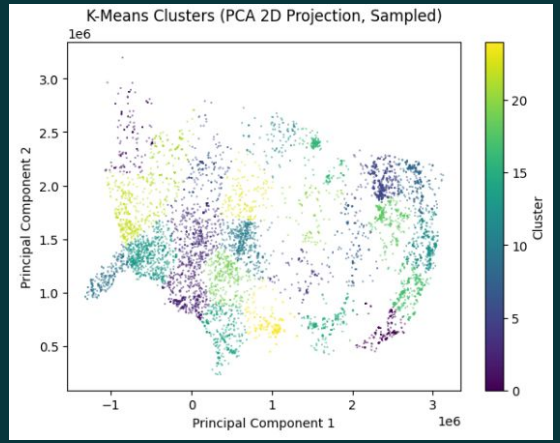
Results - K-Means

- Best k is 2 with Silhouette Score **0.7862**
- Another trial with k = 25 returned silhouette score **0.5916**



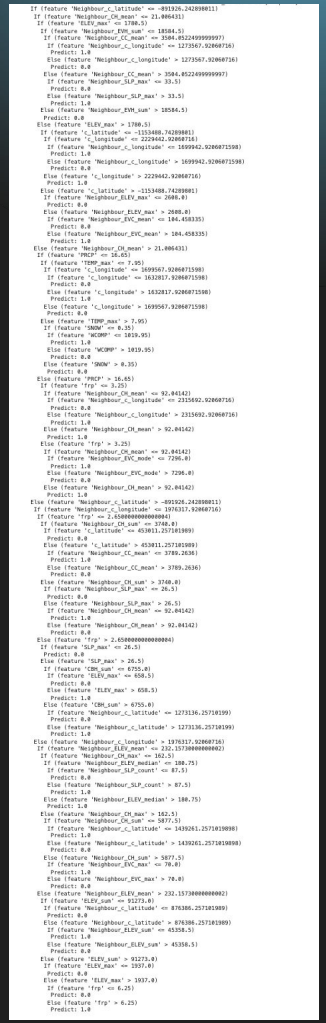






Results - Decision Tree

- Dumb (depth 0) model had accuracy of 92.2% due to class imbalance
  - 100% accuracy for fires that did not spread, 0% accuracy for fires that did spread
  - F1 score of 0
- Balanced model (depth 6)
  - Accuracy of 72.3%
  - F1 score of 0.7866
- Important decision columns
  - Latitude, canopy height, vegetation height, slope, elevation, temperature, longitude



Discussion

K-means

- Silhouette Score of **0.5916** indicates a decent clustering & that the geographic locations of the clusters can be separated fairly with some locations being a bit further from the center

- Would want to test clustering with higher K (and stronger computing power)

Decision Tree

- F1 score of 0.78 indicates real-world viability
  - 0.7 is usable while 0.8 is considered very good, depending on the task
- Varied columns represented in decision tree indicate that the model and data were a good match
- Possible improvements
  - More robust binary classification model (MLP)

Q&A

APPENDIX

```
root
 |-- acq_date: double (nullable = true)
 |-- frp: double (nullable = true)
 |-- Neighbour_frp: string (nullable = true)
 |-- ELEV_max: double (nullable = true)
 |-- ELEV_min: double (nullable = true)
 |-- ELEV_median: double (nullable = true)
 |-- ELEV_sum: double (nullable = true)
 |-- ELEV_mode: double (nullable = true)
 |-- ELEV_count: double (nullable = true)
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 |-- SLP_sum: double (nullable = true)
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```

```
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```

```
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Cluster 2: [-6.84906285e-02  4.15853540e+05  8.73321402e+05]
Cluster 3: [-6.38103343e-02  5.12266471e+05  1.46761641e+06]
Cluster 4: [ 2.06330696e-01 -8.04462021e+05  1.58780672e+06]
Cluster 5: [ 1.13191526e-01 -1.41527287e+06  2.75806280e+06]
Cluster 6: [-2.11273331e-01  6.69990868e+05  2.07554620e+06]
Cluster 7: [ 3.71479352e-01 -1.31206782e+06  2.11971614e+06]
Cluster 8: [ 8.12808134e-02 -1.82254386e+06  2.93770507e+06]
Cluster 9: [1.78233131e-02 1.39502658e+06 6.98456046e+05]
Cluster 10: [ 3.95316256e-02 -5.60908391e+04  1.60983298e+06]
Cluster 11: [ 1.32725501e-02 -6.68177373e+04  2.66768558e+06]
Cluster 12: [ 4.54803703e-02 -2.15901878e+06  2.42623812e+06]
Cluster 13: [-8.16962461e-02  9.69512370e+05  1.05372265e+06]
Cluster 14: [-1.43838544e-01 -1.12443380e+05  7.22814808e+05]
Cluster 15: [ 1.40014007e-01 -1.30621281e+06  1.29650220e+06]
Cluster 16: [-8.80549983e-02 -6.09513744e+05  2.74804785e+06]
Cluster 17: [ 1.76167228e-01 -2.12311789e+06  1.95577339e+06]
Cluster 18: [ 3.32069201e-01 -1.64236789e+06  2.48638886e+06]
Cluster 19: [-8.09701361e-02  3.49483674e+04  1.22899260e+06]
Cluster 20: [ 1.99946737e-01 -7.21550777e+05  2.18893371e+06]
Cluster 21: [-2.37601230e-01  1.08106878e+06  2.02852357e+06]
Cluster 22: [-9.41158980e-02  1.35630306e+06  1.34521280e+06]
Cluster 23: [-8.11736934e-02  8.41516711e+04  2.01227238e+06]
Cluster 24: [-1.61611375e-01 -6.38229258e+05  1.02551129e+06]
```